

DETERMINISTIC CRYPTOGRAPHIC POSTURE

SecureMail forensic report

JSON is authoritative. This HTML and the companion PDF are presentations of one in-memory canonical report. Missing evidence is not a pass.

Schema **securemail.report/v1** Generated **2026-09-05T20:04:37Z**

Profile **ietf_current** Assessment **limited**



```
Capture SHA-256
ad9f53ac6cb9d1e5a4590d40dda7eedc6a674b0576e1c1d032928fed729fca00

Working copy
present · ad9f53ac6cb9d1e5a4590d40dda7eedc6a674b0576e1c1d032928fed729fca00

Analyzer bundle
792b7a67fe8b6fe326fb9235567359d5b2d56c81ce25f4527ec8078a6f7c94c9

Policy pack
9b934ed11b792bc34f4d803d73b0098448d99cfa376074ef6560ad035cb2e9ae

Template
report.html.j2 · 9af49e7a478bad9d874034675f2268ff1d16dbc4d206b2cb20fecdeb1ff53b9d
```

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Report regions

Four regions stay separate. ML never rewrites a deterministic finding.

Observed facts

1 flows, 0 sessions, 1 handshakes, 1 certificates.

Deterministic conclusions

6 session findings, 6 endpoint findings, 20 applicable checks.

Advisory / ML

1 advisory items. Not used as compliance judgments.

Analyst conclusions

No analyst notes attached to this export.

Capture limitations

Assessment state **limited**. Unknown and not-observable checks are coverage, not a clean bill of health.

0

INCOMPLETE FLOWS

0

CONFLICTING FLOWS

0

NOT-OBSERVABLE CERTS

- Truncated packets present: **False**
- Unknown checks: **1**
- Not-observable checks: **0**

Coverage denominators

Pass / fail / unknown / not-observable always sum to applicable. Unknown is never folded into pass.

■ pass
 ■ fail
 ■ unknown
 ■ not observable

**Overall coverage**

APPLICABLE	PASS	FAIL	UNKNOWN	NOT OBSERVABLE
20	13	6	1	0

By protocol**Coverage by mail protocol**

PROTOCOL	BAR	APPLICABLE	PASS	FAIL	UNKNOWN	NOB
unclassified		20	13	6	1	0

By category**Coverage by check category**

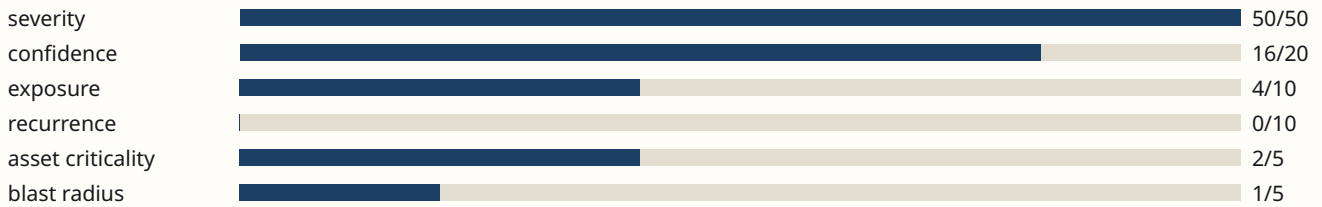
CATEGORY	APPLICABLE	PASS	FAIL	UNKNOWN	NOT OBSERVABLE
certificate	7	3	3	1	0
tls_handshake	13	10	3	0	0

Prioritized findings

CERT_PATH_INVALID_AT_CAPTURE · Certificate path invalid at capture time

high **OBS Observed** Score 73 · 172.18.0.2:4433

Offline path validation against the imported trust store failed at capture time.



Recurrence 1 · occurrences 1 · remediation pki.repair_chain

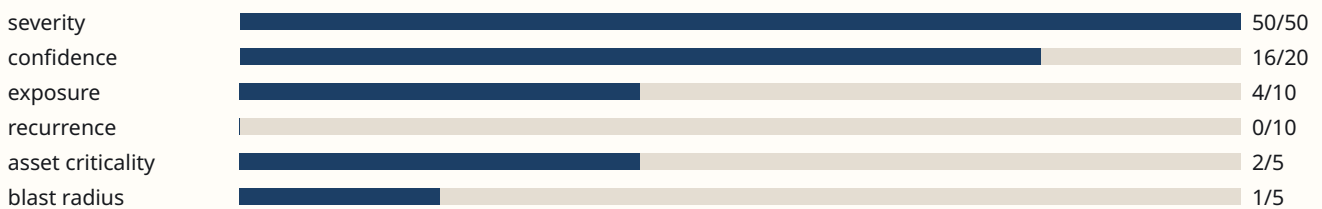
Evidence references

TYPE	KEY	FILE
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...

CERT_RSA_KEY_LT2048 · RSA authentication key shorter than 2048 bits

high **OBS Observed** Score 73 · 172.18.0.2:4433

RSA certificates used to authenticate TLS MUST use a modulus of at least 2048 bits.



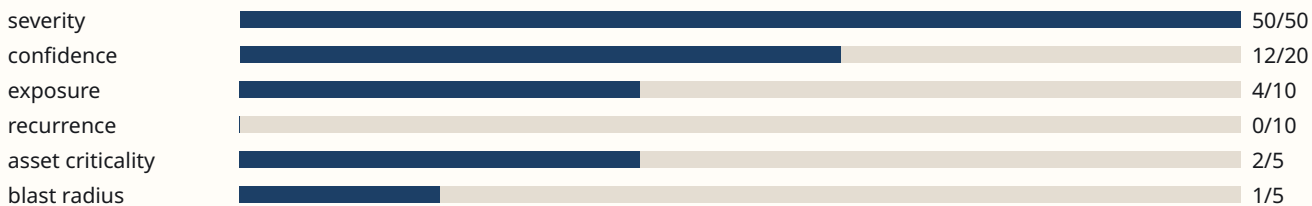
Recurrence 1 · occurrences 1 · remediation pki.rsa_2048

Evidence references

TYPE	KEY	FILE
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...
certificate	CJKFoj4bpHEhTeaRoj:server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	ca...

TLS12_STATIC_RSA_NEGOTIATED · **TLS 1.2 static RSA key exchange****high** **INF Inferred** Score 69 · 172.18.0.2:4433

Clients MUST NOT offer and servers MUST NOT select RSA key-exchange cipher suites in TLS 1.2.



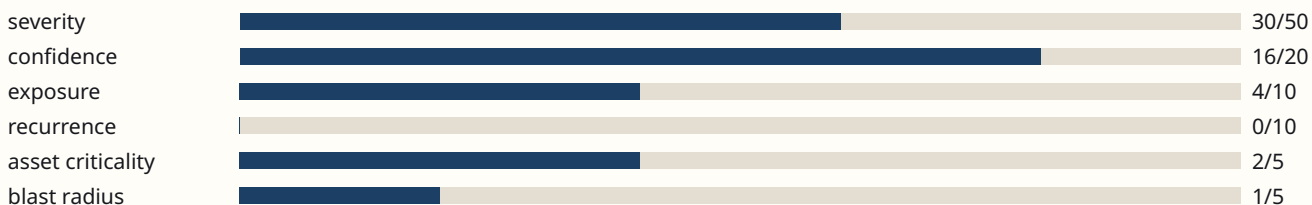
Recurrence 1 · occurrences 1 · remediation tls.disable_static_rsa

Evidence references

TYPE	KEY	FIELD	STATE	FRAME
handshake	CJKFoj4bpHEhTeaRoj	handshake.key_exchange.evidence_state	INF Inferred	—
handshake	CJKFoj4bpHEhTeaRoj	handshake.key_exchange.mechanism	INF Inferred	—
handshake	CJKFoj4bpHEhTeaRoj	handshake.version.evidence_state	OBS Observed	—
handshake	CJKFoj4bpHEhTeaRoj	handshake.version.selected	OBS Observed	—

CERT_PUBLIC_KEY_STRENGTH_LT112 · **Public key comparable strength below 112 bits****medium** **OBS Observed** Score 53 · 172.18.0.2:4433

Keys providing less than 112 bits of comparable security are below the current IETF/NIST floor.



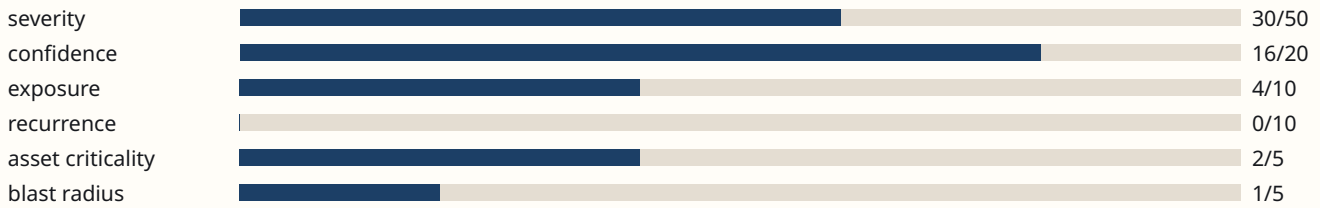
Recurrence 1 · occurrences 1 · remediation pki.minimum_strength_112

Evidence references

TYPE	KEY	FIELD
certificate	CJKFoj4bpHEhTeaRoj : server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	certificate.subjectPublicKey
certificate	CJKFoj4bpHEhTeaRoj : server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	certificate.subjectPublicKey
certificate	CJKFoj4bpHEhTeaRoj : server:0:8daee380e85c3888137c44accff505d6a7ae3227737c86b30811a4f3d7d1166e	certificate.subjectPublicKey

TLS_CIPHER_CBC · TLS CBC cipher suite negotiated**medium** **OBS Observed** Score 53 · 172.18.0.2:4433

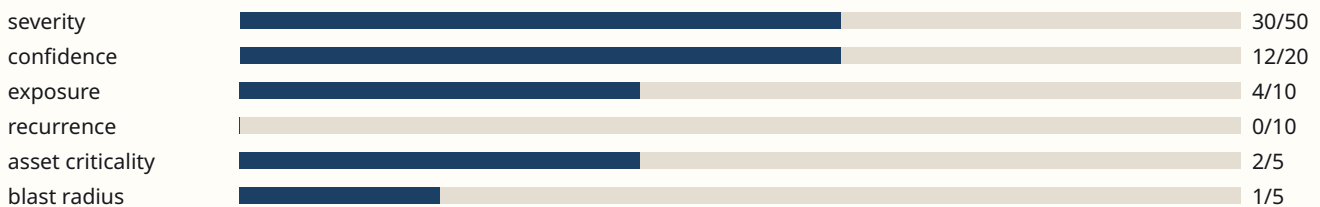
CBC cipher suites in TLS 1.2 and earlier SHOULD NOT be negotiated.

Recurrence 1 · occurrences 1 · remediation `tls.prefer_aead`**Evidence references**

TYPE	KEY	FIELD	STATE	FRAME
handshake	CJKFoj4bpHEhTeaRoj	handshake.cipher_suite.evidence_state	OBS Observed	—
handshake	CJKFoj4bpHEhTeaRoj	handshake.cipher_suite.name	OBS Observed	—
handshake	CJKFoj4bpHEhTeaRoj	handshake.version.selected	OBS Observed	—

TLS_FORWARD_SECRECY_ABSENT · Forward secrecy absent**medium** **INF Inferred** Score 49 · 172.18.0.2:4433

Static RSA, DH, or ECDH key exchange does not provide forward secrecy.

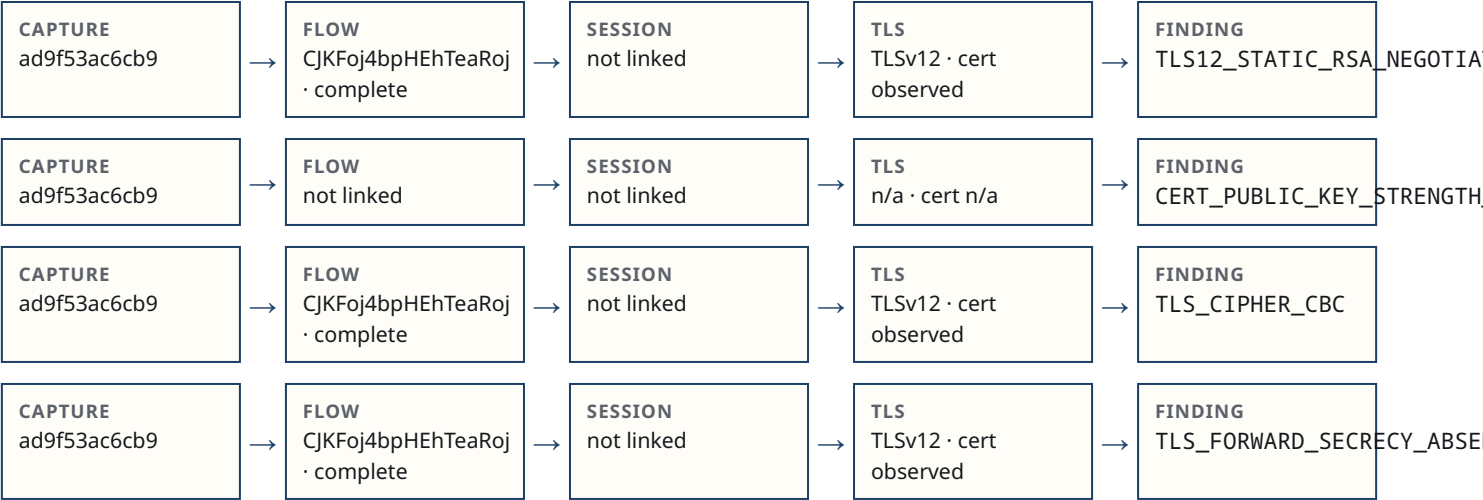
Recurrence 1 · occurrences 1 · remediation `tls.require_ephemeral_kx`**Evidence references**

TYPE	KEY	FIELD	STATE	FRAME
handshake	CJKFoj4bpHEhTeaRoj	derived.forward_secrecy.outcome	INF Inferred	—

Evidence lineage

Capture to finding. Empty hops stay labeled rather than implied.





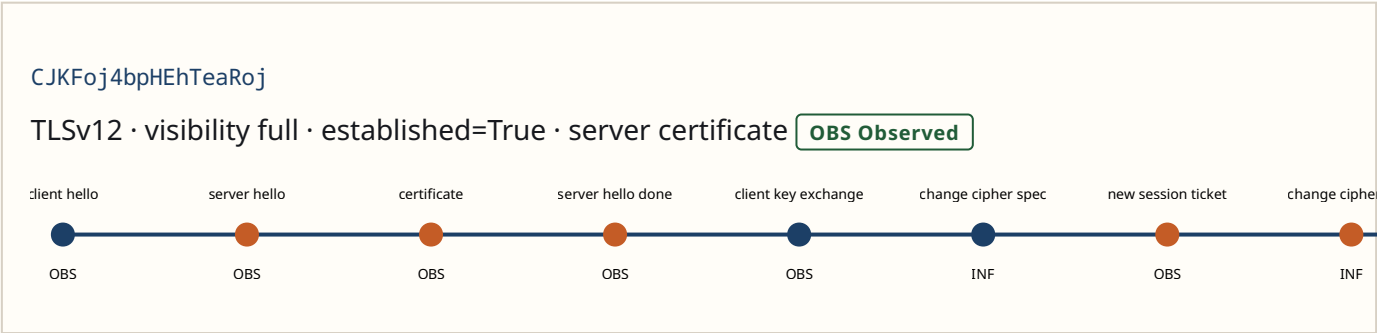
Lineage table

FINDING	ENDPOINT	FLOW	QUALITY	PROTOCOL	TLS	CERTS
CERT_PATH_INVALID_AT_CAPTURE	172.18.0.2:4433	—	—	—	—	0
CERT_RSA_KEY_LT2048	172.18.0.2:4433	—	—	—	—	0
TLS12_STATIC_RSA_NEGOTIATED	172.18.0.2:4433	CJkFoj4bpHEhTeaRoj	complete	—	TLSv12	1
CERT_PUBLIC_KEY_STRENGTH_LT112	172.18.0.2:4433	—	—	—	—	0
TLS_CIPHER_CBC	172.18.0.2:4433	CJkFoj4bpHEhTeaRoj	complete	—	TLSv12	1
TLS_FORWARD_SECRECY_ABSENT	172.18.0.2:4433	CJkFoj4bpHEhTeaRoj	complete	—	TLSv12	1

STARTTLS / STLS

No explicit-upgrade sessions in this report.

TLS handshakes



Handshake facts

UID	VERSION	CIPHER	KEX	VISIBILITY	CERT
CJkFoj4bpHEhTeaRoj	TLSv12	TLS_RSA_WITH_AES_128_CBC_SHA	RSA	full	OBS Observed

Certificate chains

CJKFoj4bpHEhTeaRoj

CHAIN_INDEX 0 · LEAF · SERVER

CN=securemail.test

Issuer CN=securemail.test · RSA 1536

OBS Observed

Observed facts

Flows

TCP reconstruction

UID	ENDPOINTS	QUALITY	REASON	STATE
CJKFoj4bpHEhTeaRoj	172.18.0.3:34612 → 172.18.0.2:4433	complete	—	OBS Observed

Sessions

Mail sessions

UID	PROTOCOL	PORT HINT	PAYLOAD	STATE	EVENTS
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Policy checks

Applicable evaluations (pass is not omitted from the denominator)

CODE	OUTCOME	STATE	PROTOCOL	CATEGORY	ENDPOINT
CERT_EXPIRED_AT_CAPTURE	pass	OBS Observed	unclassified	certificate	172.18.0.2:4433
CERT_IDENTITY_MISMATCH	unknown	IND Indeterminate	unclassified	certificate	172.18.0.2:4433
CERT_PATH_INVALID_AT_CAPTURE	fail	OBS Observed	unclassified	certificate	172.18.0.2:4433
CERT_PUBLIC_KEY_STRENGTH_LT112	fail	OBS Observed	unclassified	certificate	172.18.0.2:4433
CERT_RSA_KEY_LT2048	fail	OBS Observed	unclassified	certificate	172.18.0.2:4433
CERT_SIGNATURE_MD5	pass	OBS Observed	unclassified	certificate	172.18.0.2:4433
CERT_SIGNATURE_SHA1	pass	OBS Observed	unclassified	certificate	172.18.0.2:4433
TLS12_DHE_NEGOTIATED	pass	INF Inferred	unclassified	tls_handshake	172.18.0.2:4433
TLS12_STATIC_DH_NEGOTIATED	pass	INF Inferred	unclassified	tls_handshake	172.18.0.2:4433
TLS12_STATIC_ECDH_NEGOTIATED	pass	INF Inferred	unclassified	tls_handshake	172.18.0.2:4433
TLS12_STATIC_RSA_NEGOTIATED	fail	INF Inferred	unclassified	tls_handshake	172.18.0.2:4433
TLS_CIPHER_3DES	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_CIPHER_CBC	fail	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_CIPHER_EXPORT	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_CIPHER_NULL	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_CIPHER_RC4	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_FORWARD_SECRECY_ABSENT	fail	INF Inferred	unclassified	tls_handshake	172.18.0.2:4433
TLS_NEGOTIATED_SSL3	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_NEGOTIATED_TLS10	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433
TLS_NEGOTIATED_TLS11	pass	OBS Observed	unclassified	tls_handshake	172.18.0.2:4433

Advisory / ML

- ADVISORY_INSUFFICIENT_HISTORY — Local ML history has 1 endpoint-windows; baseline needs 14 and Isolation Forest needs 40. Deterministic findings are unchanged. field session.uid.

Analyst conclusions

No analyst notes in this export.

Provenance

Source records

KIND	IDENTIFIER	AVAILABILITY
case	capture-ad9f53ac	present
capture	ad9f53ac6cb9d1e5a4590d40dda7eedc	present
analysis_run	rd92b80d498bfaa38	present

Artifact hashes

NAME	SHA-256	AVAILABILITY
capture.pcapng	ad9f53ac6cb9d1e5a4590d40dda7eedc6a674b0576e1c1d032928fed729fca00	present

Bundled fonts

FILE	FAMILY	WEIGHT	SHA-256
NotoSans-Regular.ttf	Noto Sans	400	1f0826e3f6488a3c14fe54d0ebf860bfad1bc1cc952fd4755ff519f145c9c17e
NotoSans-Bold.ttf	Noto Sans	700	23f831d62059cc0d0cb4e71dbc9aadd1717d23c457028d81001c925e8a78cd5c
NotoSansMono-Regular.ttf	Noto Sans Mono	400	213857674181ec82bf4f825a1ddb4b2a30136ca07a30e954c229b816108fc23d

HTML renderer securemail.html/v1 · PDF weasyprint 69.0 · signature unavailable · OS/container unavailable. Canonical report digest is written beside JSON as report.json.sha256.